R Workshop – Day 3 – Exploratory Data Analysis and Statistical Graphics

Applied R for Statistical Teaching, Research and Reproducible Analysis

Contents

1 R Workshop – Day 3 – Exploratory Data Analysis and Statistical Graphics

Session 3 of 7 | Duration: 3 hours Programme: Applied R for Statistical Teaching, Research and Reproducible Analysis

1.1 Learning objectives

By the end of this session you will be able to:

1. Start from a research question and choose a chart type that answers it, rather than choosing a chart type first.
2. Explain and use the grammar of graphics: data, aesthetics, geoms, scales, labels and themes.
3. Build and interpret bar charts, histograms, box plots and scatter plots in `ggplot2`.
4. Add titles, axis labels and units so a plot stands on its own.
5. Export publication-ready graphics to a file.
6. Write a visual claim that is accurate about what the chart can and cannot show.

1.2 Capstone link

Today you produce the first visual evidence for your capstone report: three exported, interpreted graphics built from the cleaned anchor dataset (`Data/processed/anchor_dataset_clean.csv`, produced on Day 2).

1.3 Segment plan (3 hours)

Segment	Time	Purpose
Opening and recap	10 min	Reload the clean anchor dataset, recap the research question
Concept bridge	25 min	Question-led plotting and the grammar of graphics
Guided coding	60 min	Bar chart, histogram, box plot, scatter plot
Participant practice	45 min	Build and label three graphics on the anchor dataset
Interpretation and reporting	25 min	Write a two- to three-sentence interpretation per plot
Evidence log and capstone update	15 min	Export and log the three graphics

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1.4 Part 1 – Concept bridge (Core workshop activity)

1.4.1 Start with the question, not the chart

A chart type is a means, not an end. Three workshop-relevant questions and their natural chart types:

Question	Natural chart
How many lecturers are in each training track?	Bar chart of counts
How is post-training score distributed?	Histogram
Does post-training score differ by training track?	Box plot, score by track
Is post-training score related to attendance?	Scatter plot

If you find yourself with a question and no obvious chart, it is worth restating the question more precisely before opening `ggplot2`.

1.4.2 The grammar of graphics, briefly

`ggplot2` builds a plot in layers:

- **data** – the data frame.
- **aesthetics (`aes()`)** – which columns map to which visual properties (x, y, colour, fill).
- **geoms** – the visual representation (`geom_bar()`, `geom_histogram()`, `geom_boxplot()`, `geom_point()`).
- **scales** – how data values map to visual ranges (axis breaks, colour palettes).
- **labels and themes** – titles, axis labels, units, and overall appearance.

Every `ggplot2` call you write this week follows the same shape:

```
ggplot(data, aes(x = ..., y = ...)) +
  geom_...() +
  labs(title = "...", x = "...", y = "...") +
  theme_minimal()
```

1.5 Part 2 – Guided coding (Core workshop activity)

```
library(tidyverse)

lecturers <- read_csv("Data/processed/anchor_dataset_clean.csv", show_col_types =
↪  FALSE)
```

1.5.1 2.1 Bar chart – counts by training track

```
ggplot(lecturers, aes(x = training_track)) +
  geom_bar(fill = "#2C7FB8") +
  labs(
    title = "Participants by training track",
    x = "Training track", y = "Number of participants"
  ) +
  theme_minimal()
```

Interpretation example: "Online was the most common track, followed by In-Person and then Blended. This affects how much weight an Online-driven result should carry in later analysis."

1.5.2 2.2 Histogram – distribution of post-training scores

```r
ggplot(lecturers, aes(x = post_score)) +
  geom_histogram(binwidth = 5, fill = "#41AB5D", colour = "white") +
  labs(
    title = "Distribution of post-training scores",
    x = "Post-training score (0-100)", y = "Number of participants"
  ) +
  theme_minimal()
```

`binwidth` controls how fine-grained the histogram is. Try 2, 5 and 10 and discuss how the same data can look smoother or noisier purely from a plotting choice – this is itself an honesty-in-graphics point.

1.5.3 2.3 Box plot – score by training track

```r
ggplot(lecturers, aes(x = training_track, y = post_score)) +
  geom_boxplot(fill = "#FEB24C") +
  labs(
    title = "Post-training score by training track",
    x = "Training track", y = "Post-training score (0-100)"
  ) +
  theme_minimal()
```

A box plot shows the median, the interquartile range and likely outliers in one view – a natural bridge into the Day 5 question of whether track is associated with score.

1.5.4 2.4 Scatter plot – score against attendance

```r
ggplot(lecturers, aes(x = attendance, y = post_score)) +
  geom_point(alpha = 0.6, colour = "#6A51A3") +
  labs(
    title = "Post-training score by attendance",
    x = "Attendance (%)", y = "Post-training score (0-100)"
  ) +
  theme_minimal()
```

1.5.5 2.5 Export

```r
dir.create("Outputs", showWarnings = FALSE)

p_track_counts <- ggplot(lecturers, aes(x = training_track)) +
  geom_bar(fill = "#2C7FB8") +
  labs(title = "Participants by training track",
       x = "Training track", y = "Number of participants") +
  theme_minimal()

ggsave("Outputs/day3_bar_track_counts.png", p_track_counts, width = 7, height = 5,
↪  dpi = 300)
```

Always export at a fixed `width/height/dpi` so figures look the same in your report regardless of your current RStudio window size.

1.5.6 2.6 Writing a cautious visual claim

A chart shows association, not cause. Compare:

- *Overclaiming:* "Attendance increases post-training scores."
- *Accurate:* "Participants with higher attendance tended to have higher post-training scores in this sample; this plot does not establish that attendance caused the difference."

1.6 Part 3 – Participant practice (Core workshop activity)

Using `lecturers`, produce and export **three** graphics, each with a two- to three-sentence interpretation:

1. One categorical-summary chart (bar chart) for a variable of your choice (e.g. `region`, `institution_type`, `completed`).
2. One distribution chart (histogram or box plot) for `pre_score` or `post_score`.
3. One relationship chart (scatter plot) connecting two numeric variables, or a box plot connecting a category to a numeric variable, relevant to your research question.

For each: add a title, axis labels with units, export to `Outputs/`, and write your interpretation as a comment directly above the `ggsave()` call.

A facilitator-reviewed solution is in `Solution-Scripts/day3_eda_solution.R`.

1.7 Optional demonstration

- **Density plots and violin plots** – smoother alternatives to histograms/box plots; flag that both can mislead with very small groups (e.g. fewer than ~10 observations per group).
- **Two-way categorical graphics** – stacked or grouped bar charts, e.g. `completed` by `training_track`.
- **Faceting** – `facet_wrap(~ training_track)` to repeat one chart across groups.
- **Missing-data visualisation** – a simple bar chart of `colSums(is.na(lecturers))`.
- **Honest versus misleading graphics** – truncated y-axes, 3D pie charts, dual axes with mismatched scales.

```
# Optional: faceted histogram of post_score by training_track
ggplot(lecturers, aes(x = post_score)) +
  geom_histogram(binwidth = 5, fill = "#41AB5D", colour = "white") +
  facet_wrap(~ training_track) +
  labs(title = "Post-training score by training track", x = "Post-training score",
  ↪   y = "Count") +
  theme_minimal()
```

1.8 Reference or take-home material

- Theme customisation: `theme()`, `theme_bw()`, `theme_classic()`, institutional colour palettes.
- Annotation approaches: `geom_text()`, `geom_label()`, `annotate()`.

1.9 Daily deliverable

Three exported graphics from the anchor dataset (`Outputs/day3_*.png`), each with a two- to three-sentence interpretation saved alongside the script.

1.9.1 Evidence log entry

Complete the Day 3 row in `Workshop-Evidence-Log.md`: list the three graphics produced and note one early pattern in the data relevant to your research question.

1.10 Facilitator notes

- If a participant's research question does not map naturally onto bar/histogram/box plot/scatter, help them restate the question rather than forcing an awkward chart type.
- Watch for binwidth and bar-width choices made purely for aesthetics without checking whether they hide or exaggerate a pattern – this is a good moment to circle back to the "honest versus misleading graphics" discussion even if it is technically optional.
- Confirm every exported PNG actually has a title and axis labels before participants move on; an unlabelled plot is treated as incomplete for the deliverable.